

GENERAL DATA

(SERIES "TF")

ENGINE TYPE	...	...	...	...	XPAG/TF (to Chassis No. 6500).	
					XPEG/TF (from Chassis No. 6501).	
Number of cylinders	...	...	...	...	Four.	
					XPAG/TF	XPEG/TF
Capacity	...	...	...	...	1250 c.c. (76½ cu. in.)	1466 c.c. (89.4 cu. in.).
Bore	...	...	...	...	66.5 mm. (2.618 in.).	72 mm. (2.835 in.).
Stroke	...	...	...	...	90 mm. (3.543 in.).	90 mm. (3.543 in.).
Compression ratio	...	...	...	...	8.1 : 1.	8.3 : 1.
R.A.C. rating	...	...	...	...	10.97 h.p.	12.86 h.p.
B.H.P.	...	...	...	...	57 at 5,500 r.p.m.	63 at 5,000 r.p.m.
Compression ratio	...	...	...	...	8 to 1.	
System of cooling	...	...	...	...	Thermo-siphon—pump and fan assisted.	
Radiator hose top	...	...	...	...	Length 4½ in. (10.95 cm.).	Diameter 1½ in. (3.8 cm.).
Radiator hose bottom	...	...	...	...	Length 2½ in. (5.71 cm.).	Diameter 1½ in. (2.86 cm.).
First oversize bore	...	...	...	...	+0.020 in. (.50 mm.).	
Second oversize bore (maximum)	...	...	...	...	+0.040 in. (1.00 mm.).	
Firing order	...	...	...	...	1-3-4-2.	
Piston clearance at pressure face below oil ring	...	...	...	...	.0021 in. to .0029 in. (.056 mm. to .071 mm.).	
Ring gap	...	...	...	...	.006 in. to .010 in. (.152 mm. to .254 mm.).	
Number of compression rings	...	...	...	...	Two.	
Width of compression rings	...	...	...	...	2.25 mm. (.0885 in.).	
Number of oil rings	...	...	...	...	One.	
Width of oil ring	...	...	...	...	4.0 mm. (.1575 in.).	
Oil pressure relief valve operates	...	...	...	...	50 to 70 lb. per sq. in. (3.5 to 5.0 kg./cm. ²).	
Oil pressure (normal)	...	...	...	...	40 to 45 lb. per sq. in. (2.8 to 3.2 kg./cm. ²).	
Gudgeon pin type	...	...	...	...	Clamped.	
Gudgeon pin diameter	...	...	...	...	.7087 in. { +.0004 in. +.0006 in. 18 mm. { +.010 mm. +.015 mm.	
Fit in piston	...	...	...	...	Double thumb press.	
Fit in connecting rod	...	...	...	...	Clamped.	
Crankpin diameter (standard)	...	...	...	...	1.772 in. (45 mm.).	
Crankpin minimum diameter for regrind	...	...	...	...	1.722 in. (43.75 mm.).	No regrind permissible on XPEG.
Connecting rod—length between centres	...	...	...	...	7.008 in. (178 mm.).	
Connecting rod—type of bearing	...	...	...	...	Shimless, steel-backed, white-metal-lined.	
Connecting rod—side clearance	...	...	...	...	.004 in. to .006 in. (.10 mm. to .15 mm.).	
Connecting rod—diametrical clearance	...	...	...	...	.0005 in. to .002 in. (.011 mm. to .056 mm.).	
Number of crankshaft bearings	...	...	...	...	Three.	
Type of main bearings	...	...	...	...	Shimless, steel-backed, white-metal-lined.	
Standard main journal diameter	...	...	...	...	2.047 in. (52 mm.).	
Main journals first regrind diameter	...	...	...	...	2.027 in. (51.49 mm.).	No regrind permissible on XPEG.
Main journals second regrind diameter (minimum)	...	...	...	...	2.007 in. (50.98 mm.).	
Main bearings—length	...	...	...	...	Front 1.496 in. (38 mm.). Centre 1.496 in. (38 mm.). Rear 1.575 in. (40 mm.).	
Centre main bearing—end clearance	...	...	...	...	.0014 in. to .0037 in. (.035 mm. to .095 mm.).	
Main bearings—diametrical clearance	...	...	...	...	.0008 in. to .003 in. (.020 mm. to .075 mm.).	
Crankshaft—end thrust taken on	...	...	...	...	Centre bearing.	
Number of camshaft bearings	...	...	...	...	Three.	
Type of camshaft bearings	...	...	...	...	White metal (front). Zinc alloy (rear and centre).	

GENERAL DATA—continued

Camshaft—bearing clearance	...	...	...	Front .0016 in. to .004 in. (.04 mm. to .10 mm.). Rear and centre .0018 in. to .0037 in. (.045 mm. to .095 mm.).
Camshaft—end thrust taken on	...	...	...	Front end.
Camshaft—end float	...	...	...	.005 in. to .013 in. (.125 mm. to .325 mm.).
Camshaft—drive (type)	...	...	...	Duplex chain, $\frac{3}{8}$ in. pitch, 60 pitches, endless.
Valve timing markings	...	...	...	White chain links and "T" marks on wheels.
Exhaust valve throat diameter	...	...	...	1.126 in. (28.6 mm.).
Exhaust valve diameter	...	...	...	Head 1.338 in. (34 mm.); Stem .315 in. (8 mm.).
Inlet valve throat diameter	...	...	...	1.274 in. (32.6 mm.).
Inlet valve diameter	...	...	...	Head 1.417 in. (36 mm.); Stem .315 in. (8 mm.).
Valve seat angle	...	...	...	30°.
Tappet type	...	...	...	Hollow.
Valve lift—inlet	...	...	...	.315 in. (8 mm.).
Valve lift—exhaust	...	...	...	.315 in. (8 mm.).
Inlet valve opens	...	...	...	5° before T.D.C.
Inlet valve closes	...	...	...	45° after B.D.C.
Exhaust valve opens	...	...	...	45° before B.D.C.
Exhaust valve closes	...	...	...	5° after T.D.C.
Valve spring pressure—shut	...	Inner	41.1 lb. at 1.753 in. (18.64 kg. at 4.453 cm.).	Total 114.35 lb. (51.87 kg.)
	...	Outer	73.25 lb. at 1.847 in. (33.23 kg. at 4.692 cm.).	
Valve spring pressure—open	...	Inner	55 lb. at 1.438 in. (24.95 kg. at 3.678 cm.).	Total 150 lb. (68.04 kg.)
	...	Outer	95 lb. at 1.532 in. (43.09 kg. at 3.89 cm.).	
Inlet valve opens—piston traverse	...	...	...	.009 in. (.23 mm.).
Inlet valve working clearance (hot)	...	...	...	.012 in. (.30 mm.).
Exhaust valve working clearance (hot)	...	...	...	.012 in. (.30 mm.).
Valve guides	...	...	...	Removable.

FUEL SYSTEM

Fuel tank reserve level	...	...	...	Warning light on panel.
Fuel delivery	...	...	...	S.U. electric pump.
Carburettors	...	...	...	S.U. semi-down draught.
Carburettor needles	...	...	...	Standard—G.J. Rich—H.I. Weak—G.L.

CLUTCH

Type	...	...	...	...	Borg & Beck 8 in. (20.3 cm.) diameter. Dry plate.
Facing	...	...	...	...	Woven yarn type.

GEARBOX

Synchromesh	...	...	...	...	Second, third and top.
Ratios	...	...	...	...	Gearbox Overall 8/39
					Top 1.00 4.875
					Third 1.385 6.752
					Second 2.07 10.09
					Bottom 3.5 17.06
					Reverse 3.5 17.06

FRONT SUSPENSION AND STEERING

Camber (static position)	...	...	...	...	1° positive (tolerance $\pm 1^\circ$).
Castor angle	...	...	...	...	2° $\pm \frac{1}{2}^\circ$ with side-members parallel to road.
Toe-in	...	...	...	...	Nil.
King-pin inclination	...	...	...	...	9° to 10½° full bump.

GENERAL DATA—continued

Track (disc wheels)	...	...	...	...	Front 47 $\frac{3}{8}$ in. (1·203 m.). Rear 50 in. (1·27 m.).
Track (wire wheels)	...	...	...	...	Front 48 $\frac{3}{8}$ in. (1·224 m.). Rear 50 $\frac{1}{8}$ in. (1·29 m.).
Turning circle	...	...	...	...	31 ft. 3 in. (9·525 m.).
Wheelbase	...	...	...	...	7 ft. 10 in. (2·388 m.).
Tyre size	...	...	...	...	5·50—15.
Tyre pressures	...	...	...	...	18 lb. per sq. in. (1·27 kg./cm. ²).
REAR AXLE					
Type of axle	...	...	...	...	Semi-floating.
Type of drive	...	...	...	...	Hypoid.
Ratio	...	...	...	...	8/39 = 4·875 : 1. Alternative ratios 4·55 : 1 and 5·125 : 1.
Adjustment	...	...	...	...	By spacers and special fixtures.
BRAKES					
Type	...	...	...	...	Lockheed hydraulic two leading shoe (front) (9 in. dia.).
Type of linings	...	...	...	...	Ferodo DM.7.
Lining size—front	...	...	...	...	8·75 in. × 1·5 in. × 1·87 in. (22·22 cm. × 3·81 cm. × 4·7 cm.).
Lining size—rear	...	...	...	...	8·75 in. × 1·5 in. × 1·87 in. (22·22 cm. × 3·81 cm. × 4·7 cm.).
Number of rivets	...	...	...	...	Twelve per lining.
SPRINGS (front)					
Type	...	...	...	...	Coil.
Free length	...	...	...	...	9·59 in. ± $\frac{1}{8}$ in. (24·36 cm. ± 1·58 mm.).
Mean coil diameter	...	...	...	...	3·238 in. (8·24 cm.).
Number of effective coils	...	...	...	...	7·5.
Diameter of wire (ground)	...	...	...	...	·498 in. (11·27 cm.).
Maximum deflection	...	...	...	...	4·24 in. (10·78 cm.).
SPRINGS (rear)					
Type	...	...	...	...	Half-elliptic.
Length	...	...	...	...	42 $\frac{1}{2}$ in. (107·95 cm.).
Width	...	...	...	...	1 $\frac{1}{2}$ in. (3·81 cm.).
Number of leaves	...	...	...	...	Seven.
Thickness of leaves	...	...	...	...	$\frac{7}{16}$ in. (5·56 mm.).
Camber (free)	...	...	...	...	2·85 in. (7·24 cm.).
Working load	...	...	...	...	397 lb. at nil camber.
HYDRAULIC DAMPERS					
Type	...	...	...	...	Girling or Armstrong.
ELECTRICAL					
Distributor rotation	...	...	...	...	Counter-clockwise.
Automatic advance	...	...	...	...	25° on crankshaft at 3,100 r.p.m.
Contact breaker gap	...	...	...	...	·014 in. to ·016 in. (·36 mm. to ·40 mm.).
Sparking plug	...	...	...	...	Champion NA.8 (standard equipment) (14 mm.), $\frac{3}{4}$ in. reach.
Sparking plug gap	...	...	...	...	·020 in. to ·022 in. (·50 mm. to ·56 mm.).
Ignition timing	...	...	...	...	T.D.C. (full retard).
Charging system	...	...	...	...	C.V.C.
Battery	...	...	...	...	Lucas GTW9A2, 51-amp. at 10 hours. 12-volt. Positive earth return.

GENERAL DATA—continued

CAPACITIES

Sump	...	...	...	...	...	10½ pints (12.6 U.S. pints ; 5.96 litres).
Gearbox	...	...	...	...	...	1½ pints (1.5 U.S. pints ; .71 litre).
Rear axle	...	...	...	...	...	2½ pints (2.7 U.S. pints ; 1.3 litres).
Cooling system	...	...	...	...	...	10½ pints (12.3 U.S. pints ; 5.7 litres).
Brake fluid	...	...	...	...	...	1 pint (1.2 U.S. pints ; .57 litre).
Fuel tank	...	...	...	...	...	12 gallons (14.4 U.S. gallons ; 54 litres).

GENERAL DIMENSIONS

Length	...	...	...	...	...	147 in. (3.74 m.).
Width	...	...	...	...	...	59½ in. (1.518 m.).
Height	...	...	...	...	...	52½ in. (1.34 m.).
Ground clearance	...	...	...	...	...	6 in. (15.24 cm.).
Turning circle R.H.	...	...	...	...	...	31 ft. 3 in. (9.525 m.).
Turning circle L.H.	...	...	...	...	...	31 ft. 3 in. (9.525 m.).
Weight (unladen)	...	...	...	...	...	17 cwt. 1 qr. (1,932 lb.) (878 kg.).
Weight (laden and with two passengers)	...	...	...	...	...	22 cwt. (2,464 lb.) (1120 kg.).

TORQUE SPANNER DATA

Cylinder head stud nuts	...	...	...	...	600 lb./in. (6.9 kg./m.) (up to Engine No. XPAG/TF/31727).
					500 lb./in. (5.75 kg./m.) (from Engine No. XPEG/TF/31728).
Connecting rod big-end bolts (to next split pin hole)	...	...	...	...	320 lb./in. (3.7 kg./m.).
Main bearing cap nuts (to next split pin hole)	...	...	...	...	750 lb./in. (8.6 kg./m.).
Steering wheel attachment nut	...	...	...	...	500 lb./in. (5.75 kg./m.).